Task 1 – Machine Learning Classification Project

Customer Churn Prediction Using Machine Learning

1. Introduction

Customer churn prediction is a supervised machine learning problem used to identify customers who are likely to stop using a company's services.

This project develops a machine learning classification system to predict whether a telecom customer will churn based on customer account information, service details, and demographic information.

Two machine learning algorithms were implemented and compared:

- Logistic Regression
- Random Forest Classifier

The models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and 5-Fold Cross-Validation.

---

2. Objective

The objectives of this project are:

- Preprocess and clean the customer churn dataset.
- Prepare numerical and categorical features for machine learning.
- Split the dataset into training and testing sets.
- Train at least two classification algorithms.
- Compare the performance of the models.
- Evaluate the models using multiple classification metrics.
- Perform 5-Fold Cross-Validation.
- Identify important features influencing the Random Forest predictions.

---

3. Dataset

The project uses the **Telco Customer Churn dataset**.

The dataset contains information about telecom customers, including:

- Customer tenure
- Monthly charges
- Total charges
- Contract type
- Internet service
- Payment method
- Online security
- Technical support
- Senior citizen status
- Other customer service information

The target variable is:

- `Churn = Yes` → 1
- `Churn = No` → 0

Original dataset size:

7,043 rows × 21 columns

After cleaning missing values:

7,032 rows × 21 columns

---

4. Technologies Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Package Versions

- pandas 2.3.3
- numpy 2.2.6
- matplotlib 3.10.8
- seaborn 0.13.2
- scikit-learn 1.7.2

---

5. Data Preprocessing

The following preprocessing steps were performed:

1. Loaded the dataset using Pandas.
2. Converted the `TotalCharges` column to numeric values.
3. Removed rows containing invalid/missing `TotalCharges` values.
4. Removed the `customerID` column because it is an identifier and does not provide predictive information.
5. Converted the target variable `Churn` into binary values.
6. Separated numerical and categorical features.
7. Standardized numerical features using `StandardScaler`.
8. Encoded categorical features using `OneHotEncoder`.
9. Used a stratified 80:20 train-test split.

Dataset Split

Training data:

5,625 records

Testing data:

1,407 records

---

6. Machine Learning Models

6.1 Logistic Regression

Logistic Regression was used as a baseline classification algorithm.

It predicts the probability of a customer belonging to one of the two classes:

- Churn
- No Churn

6.2 Random Forest

Random Forest is an ensemble learning algorithm that combines multiple decision trees to make predictions.

It was selected as a second classification algorithm to compare its performance with Logistic Regression.

---

7. Model Evaluation

The models were evaluated using:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Test Set Results

Metric	Logistic Regression	Random Forest
Accuracy	80.38%	78.96%
Precision	64.85%	63.45%
Recall	57.22%	49.20%
F1-Score	60.80%	55.42%
ROC-AUC	0.8359	0.8140

The results show that Logistic Regression achieved higher values across the evaluated metrics on the test set.

---

8. Cross-Validation

Stratified 5-Fold Cross-Validation was performed on the training data.

Logistic Regression

Cross-validation accuracy scores:

0.8116, 0.7902, 0.7893, 0.8027, 0.8187

Mean CV Accuracy:

80.25%

Random Forest

Cross-validation accuracy scores:

0.7982, 0.7822, 0.7884, 0.7884, 0.7947

Mean CV Accuracy:

79.04%

Cross-validation was used to evaluate model consistency across different subsets of the training data.

---

9. Feature Importance

Random Forest feature importance was used to identify features that contributed most to the model's predictions.

Some of the important features were:

- Total Charges
- Tenure
- Monthly Charges
- Month-to-month Contract
- Online Security
- Electronic Check Payment Method
- Fiber Optic Internet Service
- Technical Support
- Senior Citizen status

These features provide useful information about the factors associated with customer churn predictions.

---

10. Visualizations

The project generated the following visualizations:

1. Churn Distribution
2. Model Performance Comparison
3. Logistic Regression Confusion Matrix
4. Random Forest Confusion Matrix
5. ROC Curve Comparison
6. Random Forest Feature Importance

These visualizations help interpret the dataset distribution and compare model performance.

---

11. Project Files

The project contains:

- `customer_churn.py` – Complete machine learning implementation
- `customer_churn_classification.ipynb` – Executed Jupyter Notebook
- `WA_Fn-UseC_-Telco-Customer-Churn.csv` – Dataset
- `requirements.txt` – Required Python packages and versions

- `README.md` – Project documentation
- `model_comparison.csv` – Model evaluation results
- Visualization files – Graphs and evaluation plots

---

12. How to Run the Project

Step 1: Install dependencies

```
pip install -r requirements.txt
```

Step 2: Run the Python program

```
python customer_churn.py
```

The program loads and preprocesses the dataset, trains both classification models, evaluates their performance, performs cross-validation, and generates the required plots and result files.

Step 3: Open the Notebook

Open:

```
customer_churn_classification.ipynb
```

The notebook contains the project code, outputs, metrics, and visualizations.

---

13. Conclusion

This project implemented a complete supervised machine learning classification workflow for customer churn prediction.

The workflow included data preprocessing, feature encoding and scaling, train-test splitting, Logistic Regression and Random Forest training, performance evaluation, ROC-AUC analysis, cross-validation, and feature importance analysis.

The experimental results provide a comparison of the two classification approaches and demonstrate how machine learning can be applied to customer churn prediction.

---

14. Deliverables

GitHub Repository

GitHub Repository:
[Add your GitHub repository link here]

Notebook

`customer_churn_classification.ipynb`

Python Source Code

`customer_churn.py`

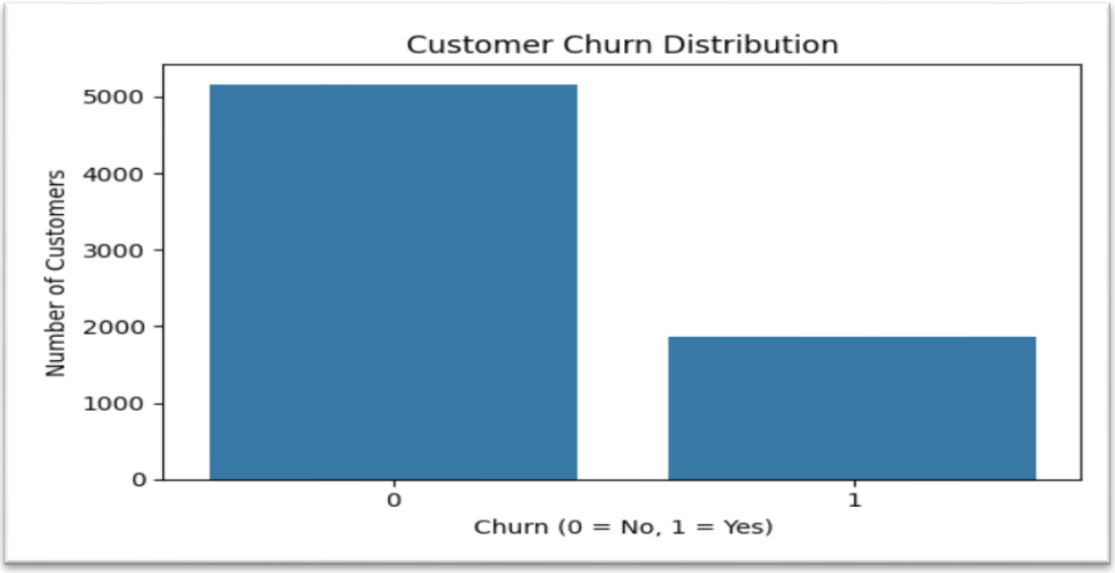
Documentation

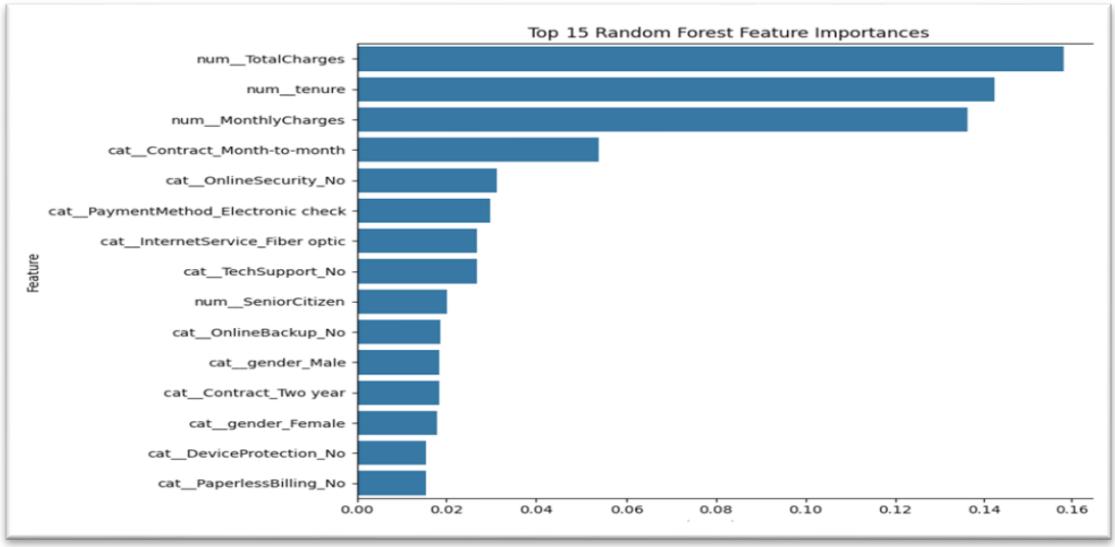
`README.md`

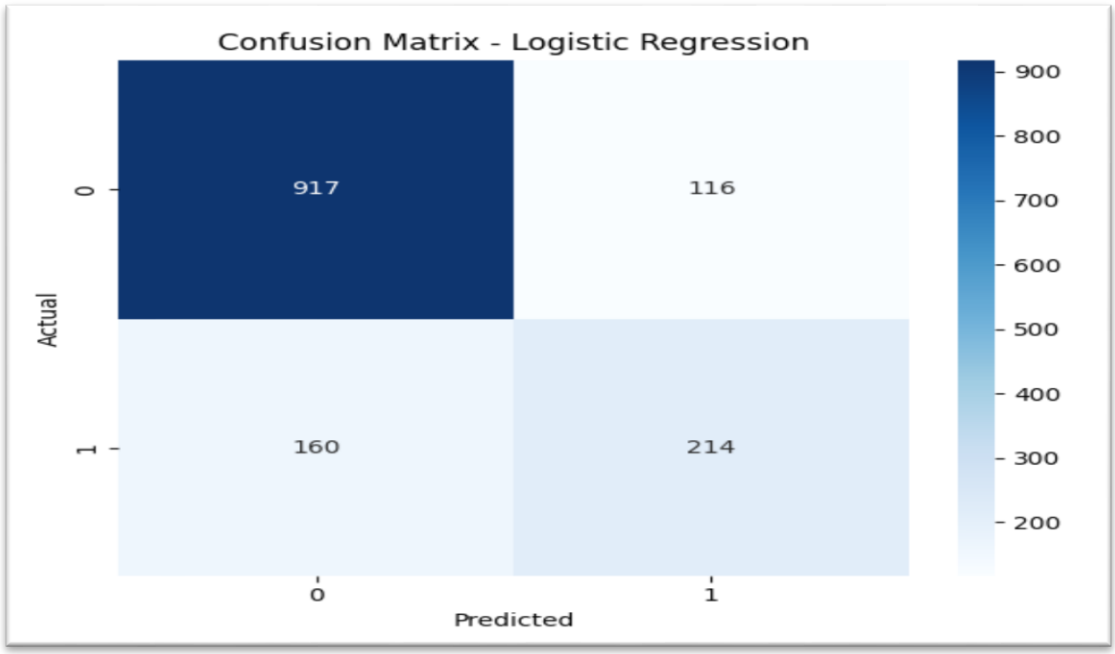
Report

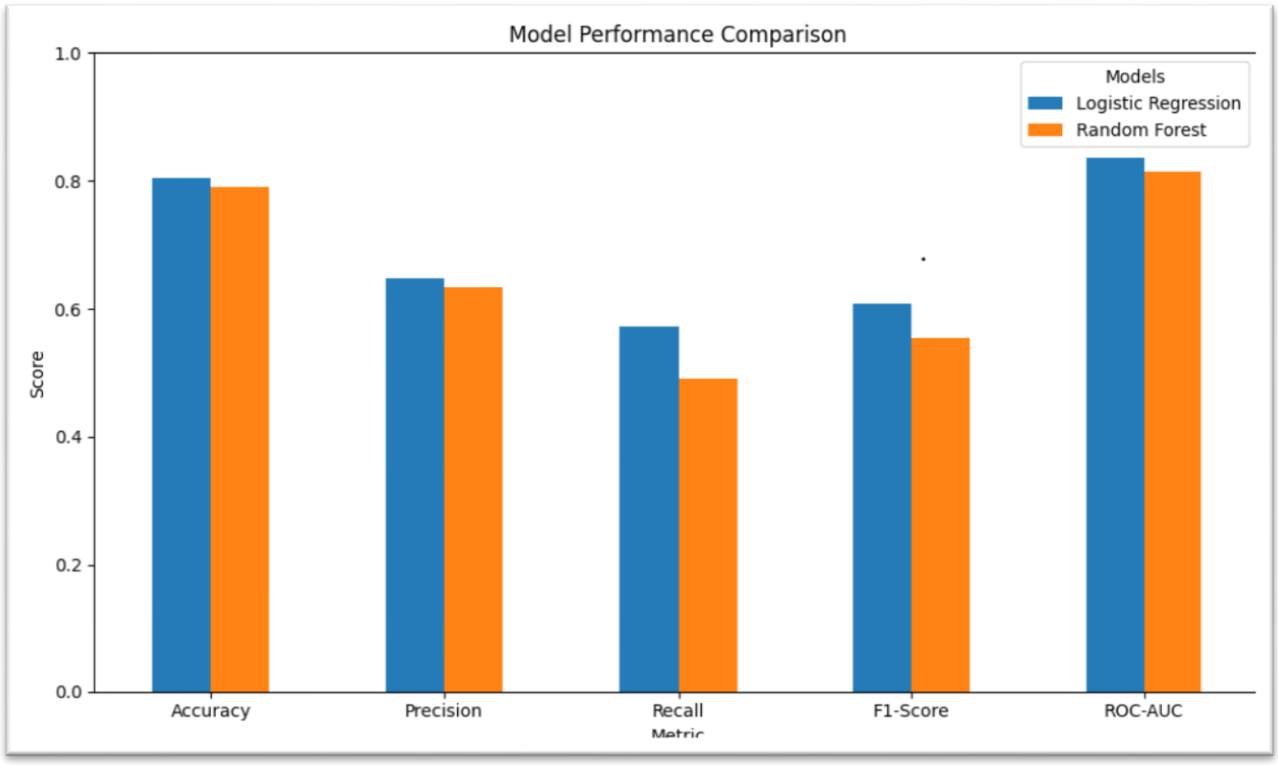
Task 1 – Machine Learning Classification Project Report

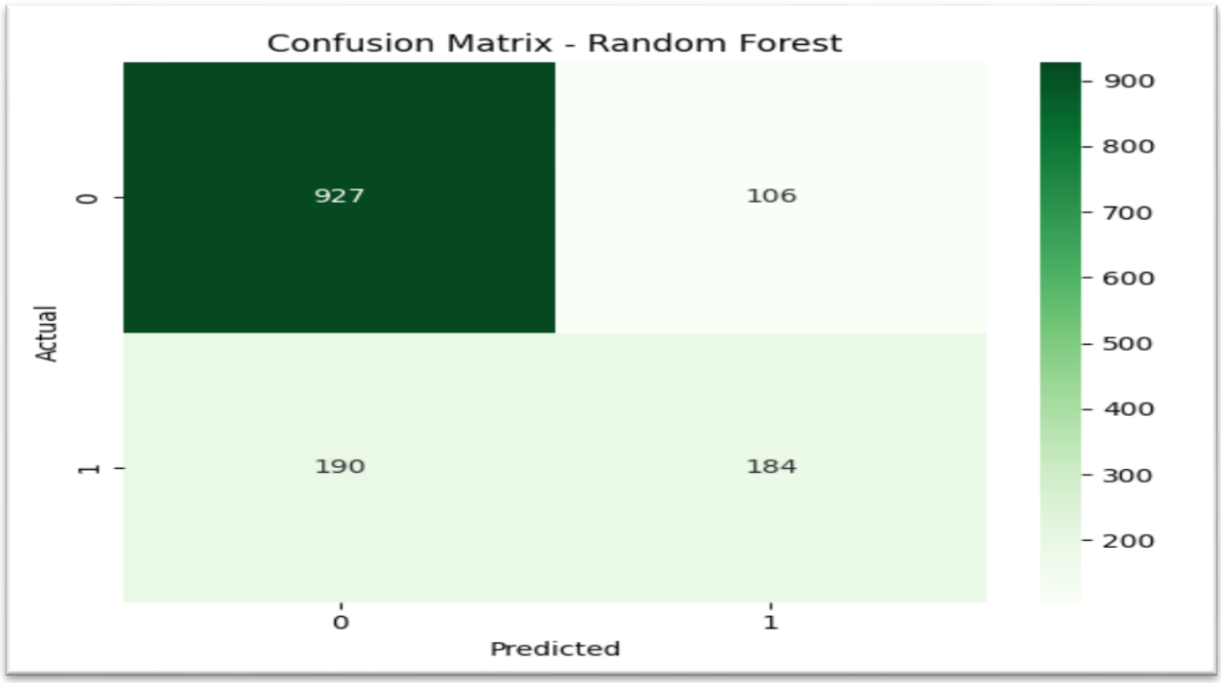
Screenshots / Visualizations

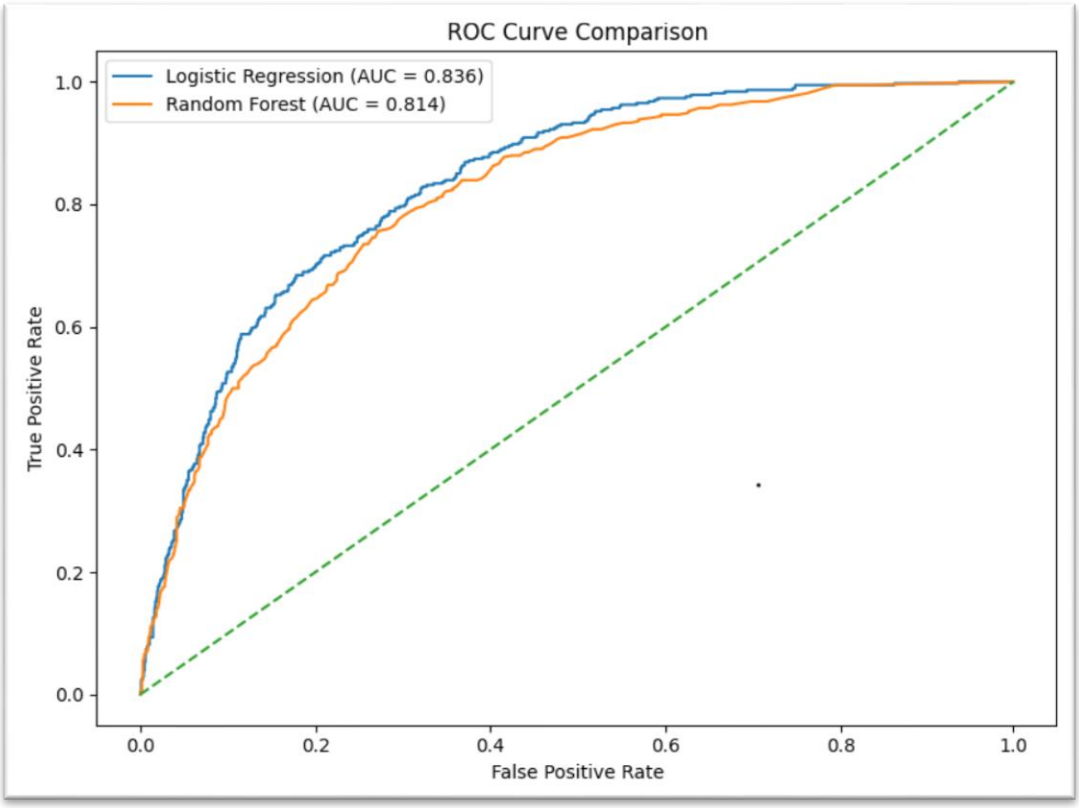



Top 15 Random Forest Feature Importances
num__TotalCharges
num__tenure
num__MonthlyCharges
cat__Contract_Month-to-month
cat__OnlineSecurity_No
cat__PaymentMethod_Electronic check
cat__InternetService_Fiber optic
cat__TechSupport_No
num__SeniorCitizen
cat__OnlineBackup_No
cat__gender_Male
cat__Contract_Two year
cat__gender_Female
cat__DeviceProtection_No
cat__PaperlessBilling_No
Feature
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16


Confusion Matrix - Logistic Regression
917
116
160
214
Actual
0
1
Predicted
0
1
900
800
700
600
500
400
300
200


Model Performance Comparison
Models
Logistic Regression
Random Forest
1.0
0.8
0.6
0.4
0.2
0.0
Score
Accuracy
Precision
Recall
F1-Score
ROC-AUC
Metric


Confusion Matrix - Random Forest
927
106
190
184
0
1
Actual
0
1
Predicted
900
800
700
600
500
400
300
200


ROC Curve Comparison
Logistic Regression (AUC = 0.836)
Random Forest (AUC = 0.814)
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0